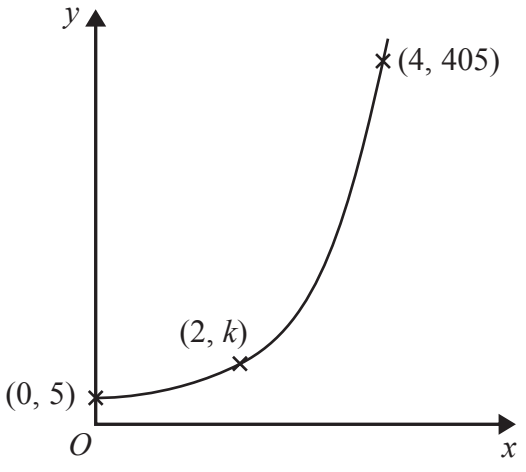
20 Here is a sketch of part of the graph of $y = pq^x$ where $q > 0$



The points (0, 5), (2, k) and (4, 405) are all on the graph of $y = pq^x$

Find the value of k.

At $x = 0$: $q^0 = 1$, so $p = 5$.

1 mark
p = 5

$$405 = 5q^4$$
$$\div 5 \qquad \div 5$$
$$81 = q^4$$

1 mark
$405 = 5q^4$

$q = 3$, since $q > 0$ and $3^4 = 81$.

1 mark
Positive fourth root

At $x = 2$: $k = 5 \times 3^2 = 45$.

1 mark
45

(Total for Question 20 is 4 marks)